CRISP: Conformal Resilient Intrusion Detection with Selective Prediction

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***Abstract*—Intrusion detection systems on Internet of Things (IoT) networks experience high class imbalance, manipulation from attackers, and lack of human supervision. In this framework, the majority of deep learning-based classifiers are not suitable for such an environment, as they provide single-label predictions with no way to abstain from predicting on uncertain inputs, defer uncertain inputs to the analyst, or detect potential unknown class attacks. We propose the CRISP (Conformal Resilient Intrusion Detection with Selective Prediction) framework, which introduces functional prediction sets where the set size reflects the ambiguity in the prediction, indicating potential unknown class attacks while preventing misclassification. Specifically, we propose a class-conditional nonconformity score algorithm that acts as an adversarial detector. To prove the validity of our framework, we present the performance on two large-scale IoT datasets: ACI-IoT-2023 with 1.2 million flows and 11 classes, and CICIDS2017 with 2.1 million flows and 8 classes, considering two different deep learning architectures, 1D-CNN and Attention-based, as backbone classifiers. Across all configurations, empirical per-class coverage remains within $\pm 0.25\%$ of the theoretical target, and within this guaranteed framework the adversarial detector attains a ROC-AUC of 0.87.**

The behaviour of the different classifiers, in the simulated adversary conditions, indicates that the effectiveness reverses with the strength of the attack: The 1D-CNN is more effective in detecting weak perturbations with an AUC score of 0.80 compared to 0.51 with the Attention-based DNN, while the Attention-based DNN is more effective with strong iterative attacks with an AUC score of 0.87 compared to 0.60 with the 1D-CNN.

This study indicates the importance of the classifier selection, as the performance is dependent on the interaction between the backbone networks' internal representation and the scoring function. Based on our knowledge, this is the first known work on the application of class-conditional conformal prediction for network intrusion detection.

***Index Terms*—Conformal prediction, intrusion detection, Internet of Things, uncertainty quantification, selective classification, adversarial robustness**

Another challenge facing the IoT infrastructure, which is expected to grow to 29 billion devices by the year 2030, is the intrusion detection issue. IoT devices include smart home devices, industrial control systems, medical devices, and even self-driving cars. IoT devices communicate with each other using different protocols such as MQTT, CoAP, Zigbee, and BLE.

Another challenge facing the IoT infrastructure is the high volume of data transmitted over the networks.

While deep learning-based intrusion detection systems have shown high accuracy of over 99% on IoT network traffic datasets, such systems have shown limited reliability in practical scenarios. Security operation center analysts have complained about the poor calibration of deep neural network-based intrusion detection systems. While the softmax output of a deep neural network may indicate a 99% confidence level for a network flow being malicious, the actual correctness of the output may be significantly less than 90% for out-of-distribution inputs. The detection of new attacks may be mapped to the most similar known category without any alert being generated. Adversarial attacks on flow-level features of network traffic are possible if an attacker is able to control IoT devices. Such attacks may cause misclassification probabilities of over 70%.

Existing uncertainty quantification methods such as MC-Dropout and Deep Ensembles provide approximate solutions but are 10–50 times more computationally expensive than the actual inference process and do not provide any finite sample guarantees for reliability. The conformal prediction (CP) method provides a different approach to uncertainty quantification. CP provides a guaranteed solution for predicting network flows and provides a set of possible outputs that is guaranteed to include the correct output with a probability of at least $1-\alpha$, where α is a significance level and a calibration set is provided. Mondrian is a variant of CP that provides a different approach to uncertainty quantification. Mondrian divides the calibration set into different classes and provides a solution for ensuring that even attacks comprising a minuscule fraction of network flows are detected.

However, conformal prediction has not yet been applied for intrusion detection in IoT systems with class-conditional guarantees, nor has its performance been evaluated in terms of adversarial evasion for such systems. We introduce a new solution, named **CRISP**, which stands for *Conformal Resilient Intrusion detection with Selective Prediction*. The research is motivated by four research questions:

- **RQ1 (Coverage):** Can Mondrian CP achieve per-class coverage within a certain tolerance for IoT data and rare classes of attacks?
- **RQ2 (Selective prediction):** Can conformal prediction sets achieve competitive or better performance compared to existing heuristic-based methods for intrusion detection while maintaining guarantees?
- **RQ3 (Adversarial detection):** Can adversarial detection be combined into a framework that also provides guarantees of coverage? What is the cost of this?
- **RQ4 (Architecture dependence):** How does the performance of adversarial detection based on a fixed conformal layer depend on the architecture of the underlying classifier?

Our main contributions are:

1) Mondrian CP is able to achieve per-class coverage within

a tolerance of $\pm 0.25\%$ of the target for two datasets, two architectures, and five significance levels (**RQ1**).
2) A unified framework for prediction, abstention, and alarm detection reaches 99.90% selective accuracy and 92.7% coverage. The prediction time is less than $4\ \mu$s (**RQ2**).
3) Adversarial detection based on a Mondrian CP framework reaches an AUC of 0.87 for PGD attacks. The cost of this is a quantified 17–19 AUC points compared to a k-Nearest Neighbors based adversarial detection (**RQ3**).
4) A complete inversion in detection performance is observed when using CNN vs. attention-based architectures (**RQ4**).

I. Related Work

Deep learning for IoT intrusion detection systems. Convolutional neural networks, long short-term memory networks, and attention-based models are widely used in this area [1]–[3], achieving over 99% accuracy on ACI-IoT-2023 [4] and CIC-IoT-2023 [5] benchmarks. Ferrag et al. [6] have reviewed over 100 deep learning-based network intrusion detection systems. These methods usually provide point predictions without any accompanying uncertainty measures or abstention functions.

Uncertainty quantification. Monte Carlo dropout [7], Bayesian convolutional neural networks [8], and similar approaches can provide uncertainty measures for deep learning-based intrusion detection systems. However, these methods are computationally expensive. Jorgensen et al. [9] have used uncertainty quantification for unknown encrypted traffic detection. However, they have not tested it for adversarial attacks. NeutroSENSE [10] uses neutrosophic logic for uncertainty quantification through shallow ensembles. SelectiveNet [11] uses a rejection function in addition to the classification function. However, it requires retraining for different coverage levels.

Adversarial robustness. Network intrusion detection systems are known to be vulnerable to adversarial attacks [12]–[15]. A2PM [16] uses flow constraints for adversarial attack detection. SHAP fingerprinting [17] detects adversarial attacks through SHAP values. Most of these methods are attack-specific or require retraining. Moreover, there are few methods that provide theoretical guarantees on detection probabilities.

Conformal prediction. Split conformal prediction [18], [19] has been successfully applied in different areas through the use of APS/RAPS scores [20], [21]. However, these methods are not widely used in network intrusion detection. Recent successful applications of these methods in network security are in concept drift detection [22] and telecommunication anomaly detection [23]. Class-conditional efficiency has been achieved through RC3P [24] for rank-calibrated class-wise coverage, clustered conformal prediction [25] for borrowing strength from similar classes, and conformal risk control [26] for group-conditional coverage under general loss functions. Mondrian conformal prediction has not been used for intrusion detection. Moreover, to the best of our knowledge, there has not been a joint evaluation of different conformal prediction methods for selective classification, adversarial attack detection, and open-set recognition in a unified framework.

II. Methodology

A. Conformal Prediction Preliminaries

Given a trained classifier $\hat{f}\colon \mathcal{X} \to \Delta^K$ with logits $z(x) \in \mathbb{R}^K$, a held-out calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, and a nonconformity score $s\colon \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, split-wise CP [18] produces prediction sets as summarized in Algorithm 1. Under exchangeability between calibration and test data, the coverage guarantee

$$P(y_{\text{test}} \in \mathcal{C}_\alpha(x_{\text{test}})) \geq 1 - \alpha \tag{1}$$

holds exactly in finite samples [27]. The ceiling formula in Step 2—rather than interpolated quantile estimation—is essential for this finite-sample property [19].

Algorithm 1 Split Conformal Prediction

Require: Classifier $\hat{f}$, calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, score $s\colon \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, level α
Ensure: $\mathcal{C}_\alpha(x_{\text{test}})$ with $P(y_{\text{test}} \in \mathcal{C}_\alpha) \geq 1 - \alpha$
1: $s_i \leftarrow s(x_i, y_i)$ for $i = 1, \ldots, n$
2: $\hat{q} \leftarrow s_{(\lceil (n+1)(1-\alpha) \rceil)}$ ▷ Conformal quantile
3: **return** $\mathcal{C}_\alpha(x_{\text{test}}) = \{y : s(x_{\text{test}}, y) \leq \hat{q}\}$

Mondrian CP stratifies calibration by class, yielding per-class quantiles $\hat{q}_k$ and the class-conditional guarantee:

$$P\big(y \in \mathcal{C}_\alpha^M(x) \mid y = k\big) \geq 1 - \alpha, \quad \forall\, k \in \mathcal{Y} \tag{2}$$

For imbalanced IoT traffic, this prevents marginal coverage from being satisfied at the expense of rare classes. With 11 classes (one left out) spanning three orders of magnitude in prevalence, marginal CP could in principle achieve 90% overall coverage while covering the rarest class at well below 50%.

B. CRISP Architecture

CRISP leverages the conformal statistics and acts as a post-hoc framework. The algorithm takes softmax outputs, logits, and a held-out calibration set as input, and operates through three modules (Fig. 1).

Module 1: Dual scoring. From a single forward pass, two nonconformity scores are computed in parallel for different operational purposes. The Adaptive Prediction Sets (APS) score [20],

$$s_{\text{APS}}(x, y) = \sum_{j=1}^{L(y)} \hat{f}(x)_{\pi_j} - (1-u) \cdot \hat{f}(x)_y \tag{3}$$

sorts softmax outputs in descending order and accumulates probability mass until the true class y is included. Here π is the descending-probability permutation, $L(y)$ is the rank of class

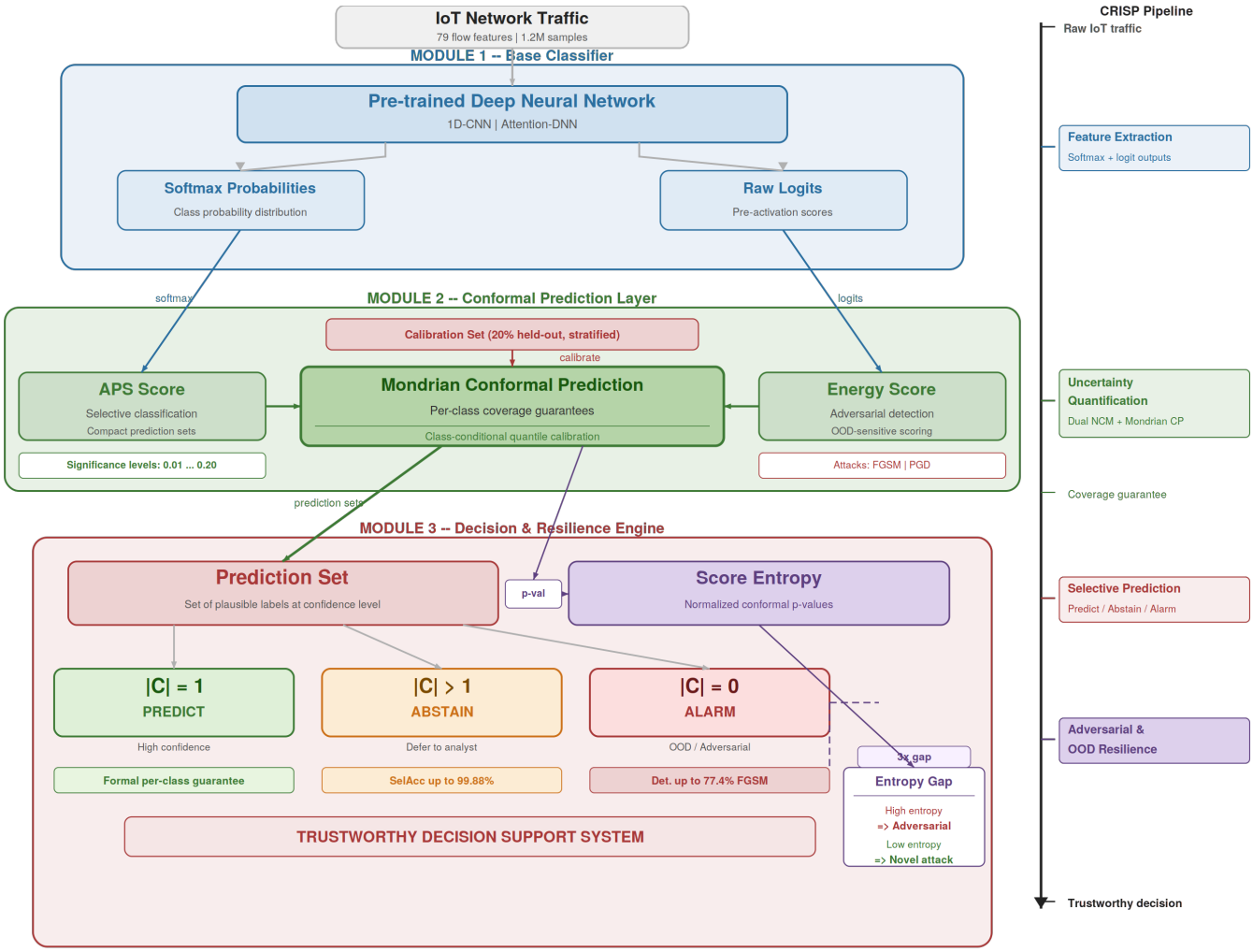

Fig. 1. Overview of the CRISP pipeline. Module 1 extracts softmax and logit representations from the base classifier. Module 2 computes APS and LogSoftmax-T nonconformity scores with Mondrian calibration. Module 3 maps set cardinality to operational decisions; conformal entropy discriminates adversarial from novel inputs.

y, and $u \sim U(0,1)$ provides randomized smoothing for finite-sample validity[1]. When the model assigns high probability to the true class, s_{APS} is small and the resulting prediction set is typically a singleton. When the model is uncertain, more classes accumulate before y is reached, producing larger sets.

The **LogSoftmax-T** score targets adversarial sensitivity:

$$s_{\text{LST}}(x, y) = -T \cdot \log \frac{e^{z_y(x)/T}}{\sum_k e^{z_k(x)/T}} \tag{4}$$

It operates on raw logits at temperature $T{=}1$ (selected by sweep over $\{0.5, 1, 2, 5\}$; see Section IV-C). However, the s_{LST} depends strictly on the hypothesised class y, limiting the viability of its standalone adoption.

[1]We implement u via SHA256 hashing of model outputs for reproducibility; collision probability $< 10^{-30}$ on `float32` vectors. Without smoothing, the non-randomized ceiling quantile $\hat{q} = s_{(\lceil (n+1)(1-\alpha) \rceil)}$ already provides valid (conservative) coverage [19]; our hash-based u tightens the bound while producing empirical coverage within $\pm 0.25\%$ of the target, consistent with the conservative baseline.

In this context, the label-free **Energy score** [28]:

$$E(x) = -T \log \sum_k e^{z_k/T} \tag{5}$$

overcome this issue assigning the same value regardless of y.

The resulting dual framework shows that the label-free scores under Mondrian calibration produced degenerate per-class quantiles, resulting in approximately 95% false alarm rates. On the other hand, the class-conditional structure of s_{LST} avoids this collapse while remaining sensitive to logit-space distortions caused by adversarial perturbation.

This study also evaluated RAPS [21] ($\lambda{=}0.01$, $k_{\text{reg}}{=}5$), which produced identical results to our approach because the base classifiers rank the true class first for over 99% of test samples, typical intrusion detection setting. Specifically, the RAPS regularization penalty effectively vanishes on data where the base model is already well-calibrated at the top of its ranking.

Module 2: Mondrian conformal layer. Unlike standard CP, which computes a single global quantile, the Mondrian variant estimates a separate quantile $\hat{q}_k$ per class from the calibration APS scores. This ensures that rare attack classes receive their own coverage threshold. At inference, the cardinality of the resulting prediction set drives a three-way operational decision: confident classification, deferral to a Tier-1 analyst, or escalation to Tier-2 investigation (Algorithm 2). The selective accuracy and coverage are:

$$\text{Sacc} = \frac{\sum_i \mathbf{1}[\hat{y}_i{=}y_i \wedge |\mathcal{C}|{=}1]}{\sum_i \mathbf{1}[|\mathcal{C}|{=}1]}, \quad \text{SCov} = \frac{\sum_i \mathbf{1}[|\mathcal{C}|{=}1]}{N} \tag{6}$$

Algorithm 2 Mondrian Conformal Layer

Require: Calibration scores $\{s_i : y_i = k\}$ for $k = 1, \ldots, K$; level α
Ensure: Operational decision for x_{test}
$\hat{q}_k \leftarrow s^{(k)}_{(\lceil (n_k+1)(1-\alpha) \rceil)}$ for each class k
$\mathcal{C}^M_\alpha \leftarrow \{y : s(x_{\text{test}}, y) \leq \hat{q}_y\}$
if $|\mathcal{C}| = 1$ **then**
  return automated response
else if $|\mathcal{C}| > 1$ **then**
  return defer to Tier-1 analyst
else
  return escalate to Tier-2 ▷ Invoke Module 3
end if

Module 3: Resilience engine. A second Mondrian CP instance, calibrated with LogSoftmax-T on the same calibration set $\mathcal{D}_{\text{cal}}$, produces per-class conformal p-values:

$$p_k(x) = \frac{|\{i \in \mathcal{D}^{(k)}_{\text{cal}} : s_i \geq s_{\text{LST}}(x, k)\}| + 1}{n_k + 1} \tag{7}$$

Reusing $\mathcal{D}_{\text{cal}}$ for both APS and LogSoftmax-T calibration is valid because each module estimates its own quantiles solely from calibration data that was never used during model training or selection; the two calibrations do not interact.

An input is flagged as anomalous when $\max_k p_k(x) < \epsilon$, where ϵ is calibrated on clean calibration data as the empirical α_{det}-quantile of $\max_k p_k$.

In our experiments, we set $\alpha_{\text{det}} = \alpha$ (the coverage level) as a separate tunable parameter; lowering α_{det} reduces false alarms at the cost of detection sensitivity. This threshold is not multiplicity-corrected across the K classes; applying Bonferroni (ϵ/K) would further reduce false alarms.

Based on our setting, our system is able to report empirical false alarm rates on clean test data directly. To distinguish adversarial inputs—uniformly nonconforming across all classes—from novel attacks that conform to some classes but not others, we compute the normalized entropy:

$$H_s(x) = -\sum_k \tilde{p}_k \log \tilde{p}_k, \quad \tilde{p}_k = p_k / \sum_j p_j \tag{8}$$

Coverage guarantees hold under exchangeability. When that assumption is violated—as with adversarial inputs or distribution drift—coverage is not assured. The conformal framework provides the detection mechanism and the detection rates are measured empirically.

III. Experimental Setup

A. Datasets

We evaluate the performance of CRISP using two benchmarks with different characteristics. The ACI-IoT-2023 dataset [4] consists of 1,231,411 flows with 79 features and 12 traffic classes in the IoT environment of a smart home. ARP Spoofing is assigned to the training set exclusively, with 5 instances. The remaining 11 classes comprise four threat families (Table I). The data distribution is uneven, with three orders of magnitude between the largest and smallest classes. The largest is Port Scan with 441K flows (35.8%), while the smallest is UDP Flood with 791 flows (0.06%). The CICIDS2017 dataset [29] consists of 2,136,413 flows with 78 features and 8 classes, excluding less frequent classes with fewer than 50 instances. The data is unevenly distributed, with 85.7% benign traffic. The least frequent class is XSS with 0.03%.

TABLE I
ACI-IoT-2023 class distribution.

Family	Class	n	%
Legitimate	Benign	329,295	26.7
Volumetric	ICMP Flood	225,234	18.3
	DNS Flood	46,935	3.8
	SYN Flood	13,857	1.1
	UDP Flood	791	0.06
Recon.	Port Scan	441,282	35.8
	Ping Sweep	71,928	5.8
	OS Scan	37,524	3.0
	Vuln. Scan	39,537	3.2
Credential	Dict. Attack	6,380	0.5
Application	Slowloris	18,643	1.5

The data is split into four disjoint stratified sets: training set (50%), validation set (10%), calibration set (20%), and test set (20%). The validation set is used only for early stopping. The calibration set is used for conformal quantile estimation but is not used at any point during model training or selection. Feature scaling is performed using the standard scaling approach, which is fit on the training set only before being applied to the data.

B. Base Classifiers

Two architecturally different models are used to assess model-agnosticism. Their structures are illustrated in Figure 2 and training details are summarized in Table II.

The two classifiers are chosen such that they cover different inductive biases. The 1D-CNN uses successive convolutional and pooling operations to capture local patterns in the feature vector depending on the position of the features.

Thus, it can efficiently recognize attacks that are represented by a group of correlated flow features.

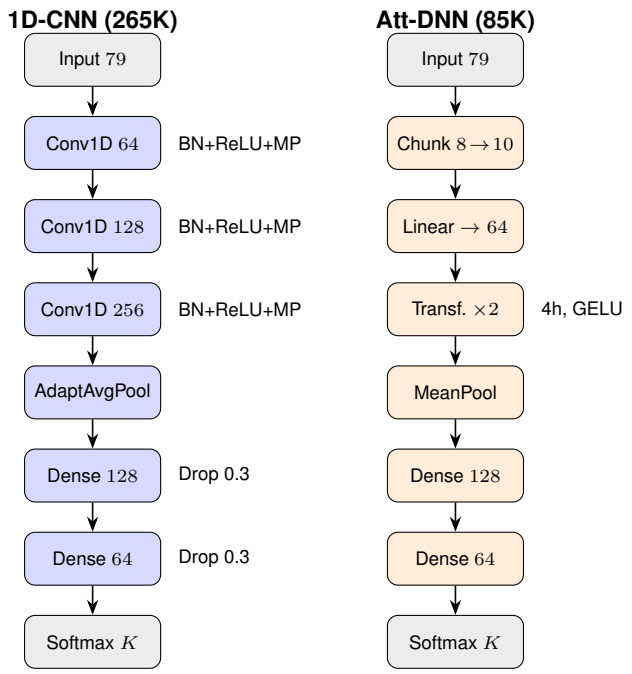

Fig. 2. Backbone architectures. BN = BatchNorm, MP = MaxPool, h = heads.

TABLE II
TRAINING CONFIGURATION AND RESULTS.

	1D-CNN	Att-DNN
Parameters	265K	85K
Optimizer	Adam [30]	
Learning rate	10^{-3}	
Weight decay	10^{-5}	
Early stopping	patience 10	
Converged epoch	73	100
Accuracy	99.55%	99.45%
F1-macro	0.970	0.955
HW: RTX 4070, 12 GB, CUDA 12.1, PyTorch 2.1		
CP overhead: $< 4\,\mu$s/sample (excl. fwd pass)		

On the other hand, the Attention-DNN utilizes the self-attention mechanism to learn global patterns in the feature vector, regardless of the feature positions. Thus, it can efficiently detect attacks that are represented by features located far from each other. The use of these two classifiers tests whether the conformal prediction guarantee of CRISP holds regardless of the representation of the input by the backbone. Finally, these two classifiers have significant differences in the number of parameters (265K vs. 85K) and the sharpness of the softmax output distributions, which directly influence the nonconformity scores and, hence, the sizes of the prediction sets. This allows us to investigate how the performance of conformal prediction, the efficiency of selective classification, and the sensitivity of adversarial attack detection depend on the representation of the input by the underlying classifier (**RQ4**).

C. Threat Model and Baselines

The threat model assumes a post-deployment adversary who has access to manipulated IoT devices and can manipulate flow-level features within a limited L_∞ budget. A white-box model is considered with respect to the base classifier, allowing the adversary to have access to gradients. This is because the adversary has no information on the calibration data or the conformal layer, implying that CRISP-based detection will not be based on security through obscurity. This is considered to be the most stringent model, implying that the detection rates should be considered as an upper bound on how an actual attacker can evade. Table III formalizes the model.

TABLE III
THREAT MODEL.

Aspect	Detail
Goal	Post-deployment evasion of IoT IDS
Capability	Modify flow features (packet sizes, timing, flags) within L_∞ budget ϵ
Knowledge	White-box w.r.t. $\hat{f}$; no access to calibration data or conformal layer
Attacks	FGSM: $\epsilon \in \{0.01, 0.05, 0.10\}$; PGD: ϵ=0.05, 20 steps, δ=0.01
Targets	Attack-class flows only; benign unperturbed
Validity	Unconstrained; detection rates are upper bounds
Open-set	UDP Flood withheld from train/calibration
Out of scope	Availability attacks on IDS

To test the performance of conformal prediction in isolation, CRISP is compared against heuristic selective classifiers and out-of-distribution detectors (Table VII). The selective baselines do not offer a guarantee of coverage in a finite sample set, while the adversarial baselines require separate detection mechanisms.

TABLE IV
BASELINES.

Task	Method	Config
Selective classif.	Softmax thr.	$\tau = 0.95$
	MC-Drop [7]	50 passes, 80th pctl
	Standard CP	Global quantile
Advers. detect.	kNN / Mahal.	k=5 / calib. logits
	Isolation For.	Softmax probs, n=200
	MSP [31]	–
	ODIN	$T \in \{100, 1000\}$

IV. RESULTS

A. Coverage Validation (RQ1)

Table V and Fig.3 display the per-class coverage results. For each significance level α, the target coverage is $1-\alpha$; for example, when $\alpha = 0.01$, each class is required to be covered at least 99% of the time. The deviation Δ_k is defined as the absolute difference between the empirical coverage on the test set and the target coverage:

$$\Delta_k = |\hat{C}_k - (1-\alpha)|, \tag{9}$$

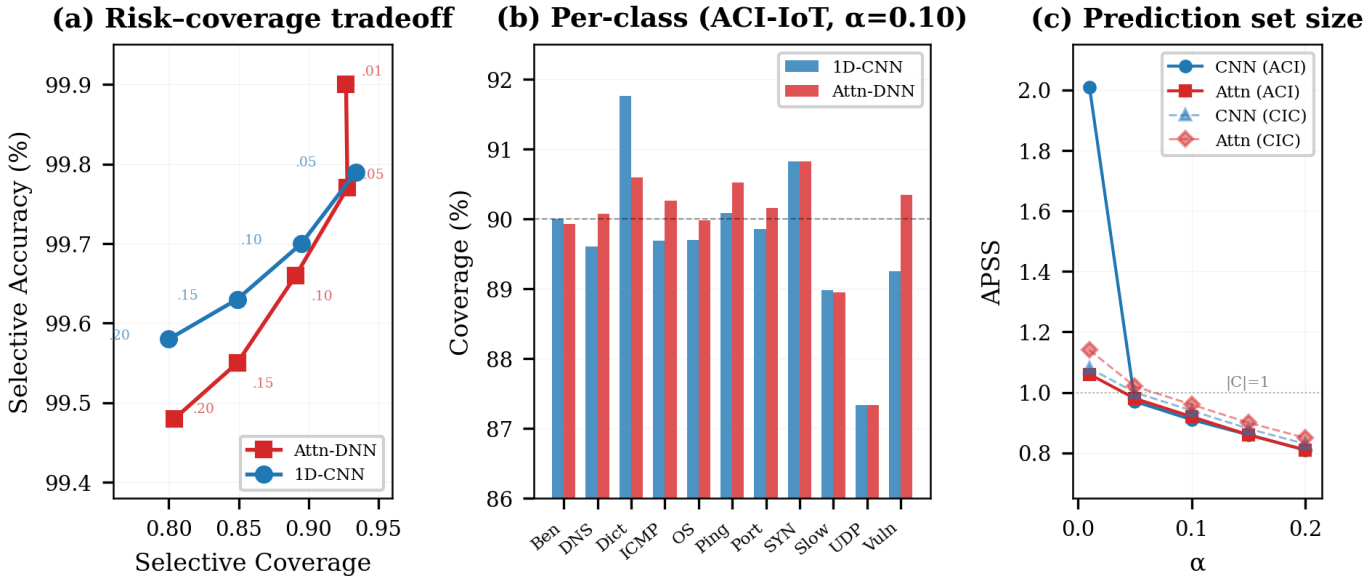

Fig. 3. (a) Risk–coverage tradeoff: CRISP traces a Pareto curve as α varies. (b) Per-class Mondrian coverage at α=0.10: all classes within $\pm$1.8% of the 90% target except UDP Flood (n=158, $-$2.7%). (c) Average Prediction Set Size across α and both datasets.

where $\hat{C}_k$ is the empirical coverage for class k. The reported $\pm 0.25\%$ corresponds to the worst-case Δ_k observed across all classes, significance levels, and architectures on ACI-IoT-2023; on CICIDS2017, the worst case is tighter at $\pm 0.12\%$. These results confirm that Mondrian quantile calibration transfers robustly from the calibration set to the test set, including for rare attack classes. On CICIDS2017, the same conformal code was applied without any dataset-specific tuning; the only difference is in the base classifier training, which uses inverse-frequency class weighting to counteract the 85.7% benign ratio.

TABLE V
COVERAGE ON ACI-IOT-2023 (246K TEST) AND CICIDS2017 (427K TEST). APSS SHOWN AS ACI/CIC.

		1D-CNN			Attn-DNN		
α	Tgt	ACI (%)	CIC (%)	APSS	ACI (%)	CIC (%)	APSS
0.01	99.0	98.98	99.01	2.01/1.08	99.02	99.01	1.06/1.14
0.05	95.0	94.86	95.04	0.97/1.00	95.08	94.98	0.98/1.02
0.10	90.0	89.85	90.04	0.91/0.94	90.13	89.98	0.92/0.96
0.15	85.0	84.89	84.99	0.86/0.88	85.18	84.96	0.86/0.90
0.20	80.0	79.98	80.01	0.81/0.83	80.25	79.88	0.81/0.85

Per-class coverage (Table VI, Fig. 3b) holds within $\pm$1.8% for every class with more than 1000 calibration samples. UDP Flood, with $n_{\text{cal}}\approx 158$, under-covers by 2.7%—the same deviation for both models, which is consistent with finite-sample quantile estimation variance rather than a model-specific deficiency. For all classes with $n_{\text{test}} > 1000$, the worst deviation is 1.05% (Slowloris).[2]

[2]As a rough guideline, the ceiling quantile $\hat{q}_k = s_{(\lceil (n_k+1)(1-\alpha) \rceil)}$ requires $n_k \geq \lceil 1/\alpha \rceil - 1$ for the index to fall within the sorted score array; for $\alpha = 0.10$, this gives $n_k \geq 9$.

Empirically, we observe stable coverage ($\Delta < 1\%$) for $n_k > 1000$ and increasing variance below that threshold.

TABLE VI
PER-CLASS COVERAGE (%) AT α=0.10 ON ACI-IOT-2023.

Class	n (test)	CNN	Attn	$\Delta_{\max}$
Benign	65,859	90.00	89.93	-0.07
DNS Flood	9,387	89.61	90.08	-0.39
Dict. Atk	1,276	91.77	90.60	$+1.77$
ICMP Flood	45,047	89.69	90.26	-0.31
OS Scan	7,505	89.70	89.98	-0.30
Ping Sweep	14,386	90.09	90.52	$+0.52$
Port Scan	88,257	89.86	90.16	-0.14
SYN Flood	2,771	90.83	90.83	$+0.83$
Slowloris	3,729	88.98	88.95	-1.05
UDP Flood	158	**87.34**	**87.34**	-2.66
Vuln. Scan	7,907	89.25	90.35	-0.75

At α=0.01, the two architectures diverge. The Attention-DNN produces near-singleton sets (APSS=1.06) with 99.90% selective accuracy at 92.7% coverage, while the CNN yields APSS=2.01—predominantly two-class sets with poor singleton accuracy. The divergence traces to softmax geometry: self-attention learns sharper logit separations, requiring less cumulative mass to reach the true class under APS. For $\alpha \geq 0.05$, both models produce APSS$<$1 and selective accuracy above 99.5%. Values below 1.0 arise from the mixture of singletons, empty sets, and occasional multi-class sets—at α=0.05, the CNN produces 93.4% singletons, 4.9% empty sets, and 1.6% multi-class sets. The risk–coverage tradeoff is shown in Fig. 3a.

B. Selective Classification (RQ2)

TABLE VII
SELECTIVE CLASSIFICATION ON CLEAN ACI-IOT-2023 DATA.

Method	Model	SelAcc (%)	SelCov (%)	Guar.
DNN (no rejection)	Attn	99.45	100	✗
Softmax τ=0.95	Attn	99.90	96.8	✗
MC-Dropout	Attn	**100.0**	80.0	✗
Standard CP α=0.10	Attn	99.65	89.3	marginal
CRISP α=0.10	Attn	99.66	89.1	✓
CRISP α=0.01	Attn	**99.90**	92.7	✓

Table VII compares selective methods on clean data. MC-Dropout reaches 100% accuracy but at a fixed 80% coverage determined by its entropy threshold, with no formal guarantee and 50× inference cost (∼200 μs per sample for 50 stochastic passes on our RTX 4070, vs. <4 μs for the conformal layer). CRISP at α=0.01 reaches the same accuracy level (99.90%) while covering 92.7% of traffic—13 percentage points more—with a formal per-class guarantee. Standard (non-Mondrian) CP achieves comparable selective accuracy to CRISP but provides only marginal coverage, which may fail to protect rare classes.

C. Adversarial Detection (RQ3) and Architecture Dependence (RQ4)

TABLE VIII
STANDALONE OOD DETECTION COMPARISON (FGSM ϵ=0.05).

Method	CNN	Attn	Mond.	Guar.
kNN logits (k=5)	**0.980**	**0.979**	✗	none
Mahalanobis	0.898	0.911	✗	none
IF softmax (n=200)	0.893	0.890	✗	none
MSP	0.874	0.843	✗	none
LogSoftmax-T (CRISP)	0.808	0.790	✓	coverage
ODIN (T=100)	0.710	0.559	✗	none

Table VIII and Fig. 4b compare detection methods. kNN on logits leads at 0.980/0.979, measuring mean Euclidean distance to the five nearest calibration neighbors in the 12-dimensional logit space. Mahalanobis (0.898/0.911) and Isolation Forest on softmax (0.893/0.890) follow. All three are label-free: they produce a single score $s(x)$ regardless of the hypothesized class, which makes them incompatible with Mondrian calibration's requirement for class-conditional scores $s(x, y)$. LogSoftmax-T is the strongest Mondrian-compatible alternative, at 0.808/0.790. The 17–19 point gap against kNN reflects the structural price of class-conditional scoring.

Considering RQ4, Fig. 4a shows a reversal in detection behavior between the two backbones. Under weak perturbation (FGSM ϵ=0.01), the CNN detects well (AUC 0.80) while the Attention-DNN is near chance (0.51). Under PGD, the Attention-DNN reaches 0.87 and the CNN drops to 0.60. Weak FGSM barely degrades the Attention-DNN's predictions—self-attention absorbs small feature perturbations without significant logit distortion—so its logits remain close to the clean distribution and escape detection. The CNN's less smooth decision boundary produces measurably distorted logits even under weak attack. PGD inverts this pattern: the CNN's flatter softmax geometry permits adversarial examples that closely mimic clean data, while the Attention-DNN's sharper logit structure produces distinct adversarial artifacts that LogSoftmax-T picks up.

A temperature sweep (Fig. 4c) confirms T=1 as optimal for both architectures. At T=0.5, detection is lower (40% CNN, 67% Attn) because the sharpened softmax suppresses score variation. At $T \geq 2$, detection collapses near zero while false alarm rates remain stable around 9%, confirming that the detection–false alarm tradeoff is controlled by score sensitivity to logit perturbations, not by threshold calibration.

D. Open-Set, Ablation, and Cross-Dataset Generalization

With UDP Flood withheld, conformal entropy separates novel from known attack traffic by 2.6× (CNN) and 3.1× (Attention-DNN). Empty-set alarm rates on the novel class are modest (4–5%), which is expected given the spectral similarity between UDP Flood and known flood variants. We note that AUROC/AUPRC for the entropy-based novel-vs-known discrimination are not reported here; the entropy gap is presented as a signal characterization rather than a calibrated detector. Operationalizing this signal—threshold selection, precision-recall characterization, and integration with the existing alarm pathway—is left for future work.

Standard and Mondrian CP produce comparable selective accuracy (99.65–99.70%), confirming that per-class calibration does not degrade clean-data performance. Switching from APS to LogSoftmax-T increases adversarial alarm rates by roughly an order of magnitude (from 6–8% to 63–73%), with the Attention-DNN benefiting more (72.6% vs. 63.1%), consistent with the architecture-dependent pattern described above.

On CICIDS2017, coverage deviates by at most ±0.12%—tighter than ACI-IoT-2023—using identical conformal code. Selective accuracy exceeds the base classifier's raw performance at every α level through principled abstention. The conformal layer requires no dataset-specific modification to transfer across traffic distributions, class counts, and imbalance ratios.

V. DISCUSSION

Coverage validity (RQ1). The discrepancy between actual and target coverage, within ±0.25% after twenty measurements, is a direct result of the conformal construction and not a random effect. The experiments verify the correctness of the implementation and the accuracy of the exchangeability. The exchangeability might not be accurate if an adversarial perturbation occurs, and coverage cannot be ensured. The detection capability offered by CRISP in such a case is empirically verified, and subsequent research focuses on its efficacy.

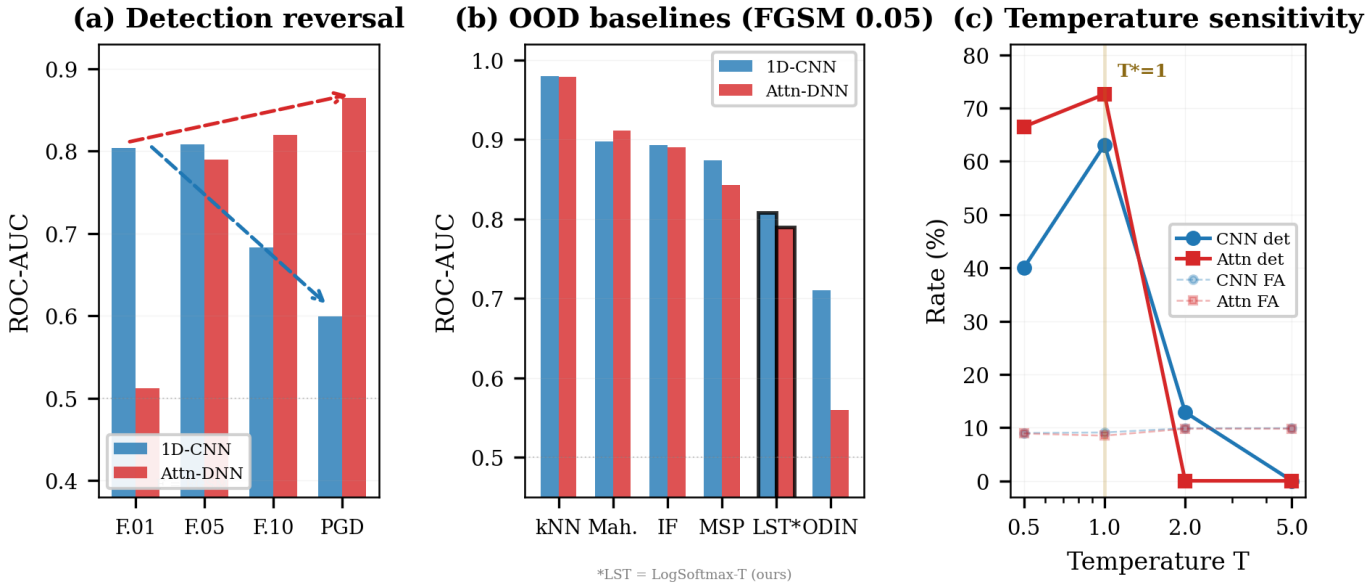

Fig. 4. (a) Detection reversal across architectures (RQ4): the CNN dominates under weak FGSM while the Attention-DNN prevails under PGD. Dashed arrows trace each model's AUC trajectory. (b) Standalone detector comparison on FGSM ϵ=0.05 (RQ3): kNN-logits at 0.98 but not Mondrian-compatible; LogSoftmax-T (black border) is the strongest compatible score. (c) Temperature sensitivity: T=1 is optimal; detection collapses at $T\geq 2$ while false alarm remains near 9%.

Selective prediction (RQ2). CRISP at $\alpha = 0.01$ matches MC-Dropout's accuracy (99.90%) at 13 percentage points more coverage, with a formal per-class guarantee instead of a heuristic threshold. In absolute terms, MC-Dropout's 50 forward passes add $\sim$200 μs per sample on our hardware, compared to CRISP's $< 4\,\mu$s conformal overhead—a 50$\times$ gap that compounds at IoT-scale throughput. The 7% of inputs deferred at $\alpha = 0.01$ represents traffic the model cannot classify with the required confidence—a fraction the operator controls through α. In a SOC workflow, this maps to the Tier-1 analyst queue, and its size can be tuned to match staffing capacity.

Integration cost (RQ3). The key trade-off of this work is the AUC point gap between 17 and 19 for LogSoftmax-T compared to kNN-on-logits. This is a direct result of Mondrian calibration, where a class-conditional score is necessary, thereby precluding label-free methods that can utilize the whole logit space. In cases where a single model is necessary to include formal coverage, abstention, and detection, none of the considered methods achieve a complete solution. For cases where detection is the principal task, kNN-on-logits is preferred because of its stronger performance on AUC, reaching 0.98, but at the cost of the formal coverage guarantees and the predict/abstain/alarm decision-making process. Possible avenues to bridge this gap include the creation of a new set of class-conditional kNN distances to be utilized as a set of nonconformity measures. The basic idea is to utilize the distance to the k-th nearest neighbor within class k itself. This approach will be compatible with Mondrian calibration of the form $s(x, y)$ while still leveraging the geometrical sensitivity of kNN to improve detection performance. Recent work on local density corrections and bagged k-distances for OOD detection [32] hints at the possibility of such a variant of kNN, at a cost of calibration complexity scaling with $K \times n_{\text{cal}}$.

Architecture effects (RQ4). The detection reversal also has direct implications for deployment. The attention-based network is better defended against iterative attackers (PGD AUC $= 0.87$), whereas the convolutional network responds more critically to low-budget single-step attacks (FGSM $\epsilon = 0.01$ AUC $= 0.80$). This indicates that even a model-agnostic framework such as ours requires a security decision regarding which model to deploy. A dual-backbone solution could be considered, where CRISP is deployed with both backbones in parallel.

VI. Limitations

There are some limitations that should be addressed. The evaluation process is based on a randomized stratified splitting approach; however, time-splitting or device-splitting might better mimic real-world scenarios because there is a high probability of correlated flows in a given time frame or from a device. The evaluation is based on adversarial perturbations; however, these may not correspond to protocol-conformant flows. It is suggested to conduct evaluations based on A2PM [16] perturbations to check the rate of detection of the proposed approach. The evaluation process is not based on adaptive attacks; these might target the proposed T-score and decision rule, which might cause a decrease in the performance of the proposed approach compared to what has been reported in this manuscript. The observed architecture-dependent reversals should be checked by using different random seeds and different architectures, including BiLSTM and/or MLP. The open-set evaluation results do not include

AUROC and AUPRC values for discrimination of novel-class; these should be computed and used to check the performance of the proposed entropy separation ratio. The proposed method has a low latency, below $4\,\mu$s, which was computed in a GPU-based setup. The evaluation of the method in a CPU-based setup, which might be used in an IoT edge device, should be performed. The performance of the method for larger values of K and calibration set sizes should be evaluated.

VII. Conclusion

The CRISP method relies on Mondrian conformal prediction to provide a post-hoc wrapper that can provide coverage guarantees, selective prediction, and adversarial detection for any classifier used in an IoT system. The coverage guarantees hold within $\pm 0.25\%$ of theoretical values in all tested scenarios (**RQ1**). The method can provide 99.9% selective accuracy and 92.7% coverage, which exceeds heuristic-based approaches and offers the advantage of formal guarantees (**RQ2**). The method can provide adversarial detection within the framework, which has a cost of 17–19 AUC points compared to standalone detectors (**RQ3**). The choice of backbone has a major, previously unreported impact on detection performance (**RQ4**).

Future work includes online CP for concept drift [22], improved rare-class set efficiency through RC3P [24] or clustered CP [25], and Mondrian-compatible scores based on class-conditional distances to close the gap with standalone detectors.

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